Homework 09, Week 12 2023.11.22 Due 2023.11.30 J20.01, J20.06, J20.12, L09.09, L09.10

★ J20.01

Demonstrate that each term of the generating functions $U_1(q, Q)$, $U_2(q, P)$, $U_3(p, Q)$ and $U_4(p, P)$ must have the units of action.

Solution

泊松括号：$[Q_i, P_j]_{\text{qp}} = \delta_{ij}$， 而 $[Q_i, P_j]_{\text{qp}} = \sum_{k=1}^{D}\left(\frac{\partial Q_i}{\partial q_k}\frac{\partial P_j}{\partial p_k} - \frac{\partial Q_i}{\partial p_k}\frac{\partial P_j}{\partial q_k}\right)$

分母是 $q_k\, p_k$，是作用量量纲，故分子也应该是作用量量纲，以保证泊松括号的结果 δ_{ij} 是无量纲的

故任何正则变换得到的 $Q_k\, P_k$，均有作用量的量纲。

对生成函数 U_1：有 $p = \frac{\partial U_1}{\partial q}$ 所以 U_1 具有 $q_k\, p_k$ 量纲，即具有作用量量纲

其它3 个生成函数类似证明。

★ J20.06

The generating function $U_2(q, P) = \sum_{k=1}^{D} g_k(q)\, P_k$ generates transformation of Lagrangian type, defined as

transformations with $Q_k = Q_k(q)$ only. Now consider a generating function $U_2(q, P) = \sum_{k=1}^{D} q_k\, f_k(P)$

(a) What condition must be imposed on the matrix with elements $\partial f_k(P)/\partial P_l$ in order for this to be a legitimate generating function?

(b) Show that in this transformation, the new momenta P_k are functions only of the old momenta p

(c) Is this transformation of the Lagrangian type? If so why, if not why not?

Solution

(a)

$$p_k = \frac{\partial U_2}{\partial q_k} \implies p_k = f_k(P) \quad \text{为构成正则变换，需要行列式} \left|\frac{\partial f_k(P)}{\partial P_j}\right| \neq 0$$

(b)

$$p_k = f_k(P) \text{ 在行列式} \left|\frac{\partial f_k(P)}{\partial P_j}\right| \neq 0 \text{ 条件下可以求得：} P_l = g_l(p) \text{ 得证}$$

(c)

作为拉格朗日类型的位形空间的点变换，需要有以下形式：$Q = h(q, t)$

现在，$Q_k = \frac{\partial U_2}{\partial P_k} = \sum_{j=1}^{D} q_j \frac{\partial f_j(P)}{\partial P_k} = \sum_{j=1}^{D} q_j\, h(P)$

代入 $P_l = g_l(p)$，得 $Q_k = \sum_{j=1}^{D} q_j\, F(p)$

Q_k 依赖于 q 和 p，不是位形空间的拉格朗日型点变换

★ J20.12 & J20.13

Consider a system consisting of a single mass with phase space variables q, $p = x, y, z, p_x, p_y, p_z$.

Let $\hat{\boldsymbol{n}}$ be an arbitrary, constant unit vector. $\vec{\boldsymbol{L}} = \vec{\boldsymbol{r}} \times \vec{\boldsymbol{p}}$ is the angular momentum vector.

(a) Show that the differential generating function （课件中称为无穷小正则变换生成元，微分生成元）

$G = \hat{\boldsymbol{n}} \cdot \vec{\boldsymbol{p}}$ generating a translation $\delta \vec{\boldsymbol{r}} = \epsilon \hat{\boldsymbol{n}}$ with no change in $\vec{\boldsymbol{p}}$.

(*b*) Show that the differential generating function $G = -\hat{\boldsymbol{n}} \cdot \vec{\boldsymbol{r}}$ generates a change of momentum $\delta \vec{\boldsymbol{p}} = \epsilon \hat{\boldsymbol{n}}$ with no change in position.

(*c*) Show that the differential generating function $G = \hat{\boldsymbol{n}} \cdot \vec{\boldsymbol{L}}$ generates changes of position and momentum

$\delta \vec{\boldsymbol{r}} = \epsilon \hat{\boldsymbol{n}} \times \vec{\boldsymbol{r}}, \quad \delta \vec{\boldsymbol{p}} = \epsilon \hat{\boldsymbol{n}} \times \vec{\boldsymbol{p}}$

(*d*) Show that the change given in step (*c*) denotes a differential rotation of $\vec{\boldsymbol{r}}$ and $\vec{\boldsymbol{p}}$ by angle ϵ about axis $\vec{\boldsymbol{n}}$

Solution

(*a*)

$G = \hat{\boldsymbol{n}} \cdot \vec{\boldsymbol{p}}$ 作为无穷小正则变换生成元时

$$\delta \vec{\boldsymbol{r}} = \vec{\boldsymbol{R}} - \vec{\boldsymbol{r}} = \epsilon \frac{\partial G}{\partial \vec{\boldsymbol{p}}} = \epsilon \frac{\partial (\hat{\boldsymbol{n}} \cdot \vec{\boldsymbol{p}})}{\partial \vec{\boldsymbol{p}}} = \epsilon \hat{\boldsymbol{n}}$$

$$-\delta \vec{\boldsymbol{p}} = \vec{\boldsymbol{p}} - \vec{\boldsymbol{P}} = \epsilon \frac{\partial G}{\partial \vec{\boldsymbol{q}}} = 0 \implies \vec{\boldsymbol{P}} = \vec{\boldsymbol{p}}$$

(*b*)

$G = -\hat{\boldsymbol{n}} \cdot \vec{\boldsymbol{r}}$ 作为无穷小正则变换生成元时

$$\delta \vec{\boldsymbol{r}} = \vec{\boldsymbol{R}} - \vec{\boldsymbol{r}} = \epsilon \frac{\partial G}{\partial \vec{\boldsymbol{p}}} = 0 \implies \vec{\boldsymbol{R}} = \vec{\boldsymbol{r}}$$

$$-\delta \vec{\boldsymbol{p}} = \vec{\boldsymbol{p}} - \vec{\boldsymbol{P}} = \epsilon \frac{\partial G}{\partial \vec{\boldsymbol{q}}} = -\epsilon \frac{\partial (\hat{\boldsymbol{n}} \cdot \vec{\boldsymbol{r}})}{\partial \vec{\boldsymbol{r}}} = -\epsilon \vec{\boldsymbol{n}} \implies \delta \vec{\boldsymbol{p}} = \epsilon \vec{\boldsymbol{n}}$$

(*c*)

$G = \hat{\boldsymbol{n}} \cdot (\vec{\boldsymbol{r}} \times \vec{\boldsymbol{p}})$ 作为无穷小正则变换生成元时

$$\delta \vec{\boldsymbol{r}} = \vec{\boldsymbol{R}} - \vec{\boldsymbol{r}} = \epsilon \frac{\partial G}{\partial \vec{\boldsymbol{p}}} = \epsilon \frac{\partial [\hat{\boldsymbol{n}} \cdot (\vec{\boldsymbol{r}} \times \vec{\boldsymbol{p}})]}{\partial \vec{\boldsymbol{p}}} = \epsilon \frac{\partial [(\hat{\boldsymbol{n}} \times \vec{\boldsymbol{r}}) \cdot \vec{\boldsymbol{p}})]}{\partial \vec{\boldsymbol{p}}} = \epsilon \hat{\boldsymbol{n}} \times \vec{\boldsymbol{r}}$$

$$-\delta \vec{\boldsymbol{p}} = \vec{\boldsymbol{p}} - \vec{\boldsymbol{P}} = \epsilon \frac{\partial G}{\partial \vec{\boldsymbol{q}}} = \epsilon \frac{\partial [\hat{\boldsymbol{n}} \cdot (\vec{\boldsymbol{r}} \times \vec{\boldsymbol{p}})]}{\partial \vec{\boldsymbol{r}}} = -\epsilon \frac{\partial [(\vec{\boldsymbol{n}} \times \vec{\boldsymbol{p}}) \cdot \vec{\boldsymbol{r}}]}{\partial \vec{\boldsymbol{r}}} = -\epsilon \vec{\boldsymbol{n}} \times \vec{\boldsymbol{p}} \implies \delta \vec{\boldsymbol{p}} = \epsilon \vec{\boldsymbol{n}} \times \vec{\boldsymbol{p}}$$

(*d*)

右图是矢量 $\vec{\boldsymbol{r}}$ 绕 $\hat{\boldsymbol{n}}$ 转 ϵ

矢量 $\vec{\boldsymbol{r}}$ 变为 $\vec{\boldsymbol{r}}'$，$\vec{\boldsymbol{r}}' - \vec{\boldsymbol{r}}$ 为红色 $\delta \vec{\boldsymbol{r}}$

先看 $\delta \vec{\boldsymbol{r}}$ 的大小：

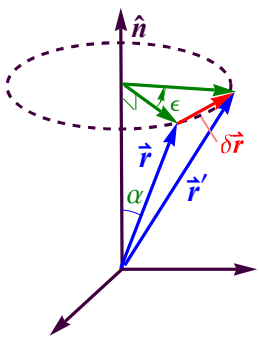
$$|\delta \vec{\boldsymbol{r}}| = |\vec{\boldsymbol{r}}' - \vec{\boldsymbol{r}}| = (r \sin \alpha) \epsilon = |(\vec{\boldsymbol{n}} \times \vec{\boldsymbol{r}}) \epsilon|$$

再确定方向：

由 图 可知，$\delta \vec{\boldsymbol{r}}$ 的方向正是 $\hat{\boldsymbol{n}} \times \vec{\boldsymbol{r}}$ 方向

写成 矢量形式，即

$$\delta \vec{\boldsymbol{r}} = \vec{\boldsymbol{r}}' - \vec{\boldsymbol{r}} = \epsilon \vec{\boldsymbol{n}} \times \vec{\boldsymbol{r}}$$



从而：任意 矢量 $\vec{\boldsymbol{r}}$ 绕 $\hat{\boldsymbol{n}}$ 转过无穷小角度 ϵ

矢量的变化量为 $\delta \vec{\boldsymbol{r}} = \epsilon \vec{\boldsymbol{n}} \times \vec{\boldsymbol{r}}$

★ L09.09

已知生成函数 $U(q, P, t) = (q_1 \cos \omega t + q_2 \sin \omega t) P_1 + (-q_1 \sin \omega t + q_2 \cos \omega t) P_2 + q_3 P_3$，求正则变换。

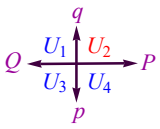
如果哈密顿函数 $H = \frac{1}{2m}\left(p_1^2 + p_2^2 + p_3^2\right) + V(r)$，其中 $r^2 = q_1^2 + q_2^2 + q_3^2$，求变换后的哈密顿函数 $K(Q, P, t)$

Solution

U 为第二类 生成函数，如图。

从而：

$$p = \frac{\partial U}{\partial q}, \quad Q = \frac{\partial U}{\partial P}$$



得：$\begin{cases} p_1 = \partial U/\partial q_1 = P_1 \cos\omega t - P_2 \sin\omega t \\ p_2 = \partial U/\partial q_2 = P_1 \sin\omega t + P_2 \cos\omega t \\ p_3 = \partial U/\partial q_3 = P_3 \end{cases}$ (†) $\quad \begin{cases} Q_1 = \partial U/\partial P_1 = q_1 \cos\omega t + q_2 \sin\omega t \\ Q_2 = \partial U/\partial P_2 = -q_1 \sin\omega t + q_2 \cos\omega t \\ Q_3 = \partial U/\partial P_3 = q_3 \end{cases}$ (#)

由式 (#) 得：$\begin{cases} q_1 = Q_1 \cos\omega t - Q_2 \sin\omega t \\ q_2 = Q_1 \sin\omega t + Q_2 \cos\omega t \\ q_3 = Q_3 \end{cases}$ (‡) 正则变换为 式 (†) 和 式 (‡)

以 式 (†) 和 式 (‡) 代入哈密顿量（哈密顿函数），并注意到 $K - H = \frac{\partial U}{\partial t}$，可得

$$K = H + \frac{\partial U}{\partial t} = \frac{1}{2m}\left(P_1^2 + P_2^2 + P_3^2\right) - \omega(Q_1 P_2 - Q_2 P_1) + V(R), \quad \text{其中 } R^2 = Q_1^2 + Q_2^2 + Q_3^2 = q_1^2 + q_2^2 + q_3^2 = r^2$$

此正则变换为点变换，新坐标绕原坐标的 q_3 轴以角速度 ω 旋转。

由原哈密顿量可判断，原坐标系为惯性系。新坐标系为非惯性系

★ L09.10

设系统的哈密顿量 $H = \frac{1}{2}\left(X q^2 + 2 Y q p + Z p^2\right)$，其中$q$，$p$ 分别为系统的广义坐标与广义动量，(X, Y, Z) 是常参数且 $X Z > Y^2$。

(i) 证明系统作简谐振动，振动频率为 $\omega = \sqrt{X Z - Y^2}$

(ii) 求出作用量变量 J，并证明它可以表示为 $J = \frac{2\pi E}{\omega}$，其中 E 为系统的机械能

(iii) 做变换 $q = \sqrt{Z}\, Q$ 且变换是正则的，变换后的系统哈密顿量为：$K = \frac{1}{2} P^2 + \frac{1}{2}\omega^2 Q^2$，求出 P 与 q，p 的关系以及生成函数。

Solution

(i)

哈密顿方程：$\begin{cases} \dot{q} = \partial H/\partial p = Y q + Z p & (†) \\ \dot{p} = -\partial H/\partial q = -X q - Y p & (‡) \end{cases}$

由 (†) 式 得：$p = \left(\dot{q} - Y q\right)/Z$ (#)

代入 (‡) 式 得：$\ddot{q} + \left(X Z - Y^2\right) q = 0$

解得：$q = q_0 \cos(\omega t + \phi)$ (★)

其中振动角频率 $\omega = \sqrt{X Z - Y^2}$

另解：

哈密顿量 $H = \frac{1}{2}\left(X q^2 + 2 Y q p + Z p^2\right)$ 不显含时间，是守恒量，$H = E$

相空间曲线：$\frac{X}{2E} q^2 + \frac{Y}{E} q p + \frac{Z}{2E} p^2 = 1$ (♈)

利用二次方程 $a x^2 + 2 b x y + c y^2 = 1$ 在 $b^2 < a c$ 时是个面积为 $\frac{\pi}{\sqrt{a c - b^2}}$ 的椭圆

故 (♈) 式的椭圆面积是：$A = \frac{2\pi E}{\sqrt{X Z - Y^2}}$

$$J = \frac{1}{2\pi}\oint p\,dq = \frac{A}{2\pi} = \frac{1}{2\pi}\frac{2\pi E}{\sqrt{XZ - Y^2}} \quad \Longrightarrow \quad E = J\sqrt{XZ - Y^2}$$

$$\omega = \frac{\partial E}{\partial J} = \sqrt{XZ - Y^2} \quad \Longrightarrow \quad J = \frac{E}{\omega}$$

(ii)

以式 (#) 及 式 (⋆) 代入 H，可得 $H = \frac{1}{2}\left(X q^2 + 2 Y q p + Z p^2\right) = \frac{\omega^2 q_0^2}{2Z} = E$

作用量变量

$$J = \oint p\,dq = -\oint \frac{(\dot{q} - Y q)}{Z}\,\omega q_0 \sin(\omega t + \phi)\,dt$$

$$= \frac{\omega^2 q_0^2}{2Z}\int_0^{2\pi/\omega}\left[1 - \cos 2(\omega t + \phi) + \frac{Y}{\omega}\sin 2(\omega t + \phi)\right]dt = \frac{2\pi E}{\omega}$$

(iii)

因为哈密顿量不显含时间，可假设变换不显含时间

$$q = \sqrt{Z}\,Q \qquad 2H = \left(XZQ^2 + 2Y\sqrt{Z}\,Q p + Z p^2\right) = 2K = P^2 + \omega^2 Q^2 = P^2 + \left(XZ - Y^2\right)Q^2$$

$$\Longrightarrow \quad P^2 = 2Y\sqrt{Z}\,Q p + Z p^2 + Y^2 Q^2 = \left(YQ + \sqrt{Z}\,p\right)^2$$

因为 q, P 都可以表示为 Q, p 的函数，故可选 Q, p 为自变量

$$P = YQ + \sqrt{Z}\,p$$

$$dU = -q\,dp - P\,dQ = \sqrt{Z}\,Q\,dp - \left(YQ + \sqrt{Z}\,p\right)dQ$$

积分得：$U = -\frac{Y}{2}Q^2 - \sqrt{Z}\,p Q, \quad P = \frac{Y q + Z p}{\sqrt{Z}}$